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Dop: 24/07/2026

EXPERIMENT NUMBER: 3

AIM: .
To study and simulate the concept of frequency reuse in cellular systems using MATLAB, by generating a hexagonal cellular layout, grouping the cells into a cluster of size N, assigning/reusing a limited set of frequency channels across the cells, and visualizing the resulting frequency reuse pattern.

OBJECTIVES:

To study the concept of frequency reuse in a cellular communication system. To understand the hexagonal cell structure used in cellular networks. To demonstrate the assignment of different frequency groups to adjacent cells.

To observe the reuse of the same frequencies in non-adjacent cells without causing significant interference

THEORY:

The Cellular Concept :

In cellular mobile communication, the total geographic coverage area is divided into small regions called cells, each served by its own Base Transceiver Station (BTS) with a limited number of frequency channels. Cells are conventionally modeled as regular hexagons, since a hexagonal grid closely approximates a circular coverage area while tiling the plane without gaps or overlaps.

Why Frequency Reuse is Needed:

The radio frequency spectrum allocated to a cellular operator is limited. It is not possible to assign a unique, nonoverlapping frequency channel to every cell in a large network. Frequency reuse solves this problem: the same set of frequency channels is reused in different cells, provided those cells are separated by a sufficient distance so that co-channel interference (interference between cells using the same frequency) remains within an acceptable limit.

Cluster Size (N)

A group of adjacent cells that together use the entire available set of frequency channels exactly once is called a cluster.

The number of cells in a cluster, N, is called the cluster size, and it must satisfy the geometric relation:

Valid Cluster Sizes $N = i^2 + i \cdot j + j^2$

(i, j are non-negative integers, not both zero)

This formula only allows certain valid values of N, namely: 1, 3, 4, 7, 9, 12, 13, 16, 19, 21 …

The entire cluster pattern is then repeated (tiled) across the whole coverage area, so every cell belongs to exactly one of the N frequency groups.

Co-channel Reuse Ratio Cells:

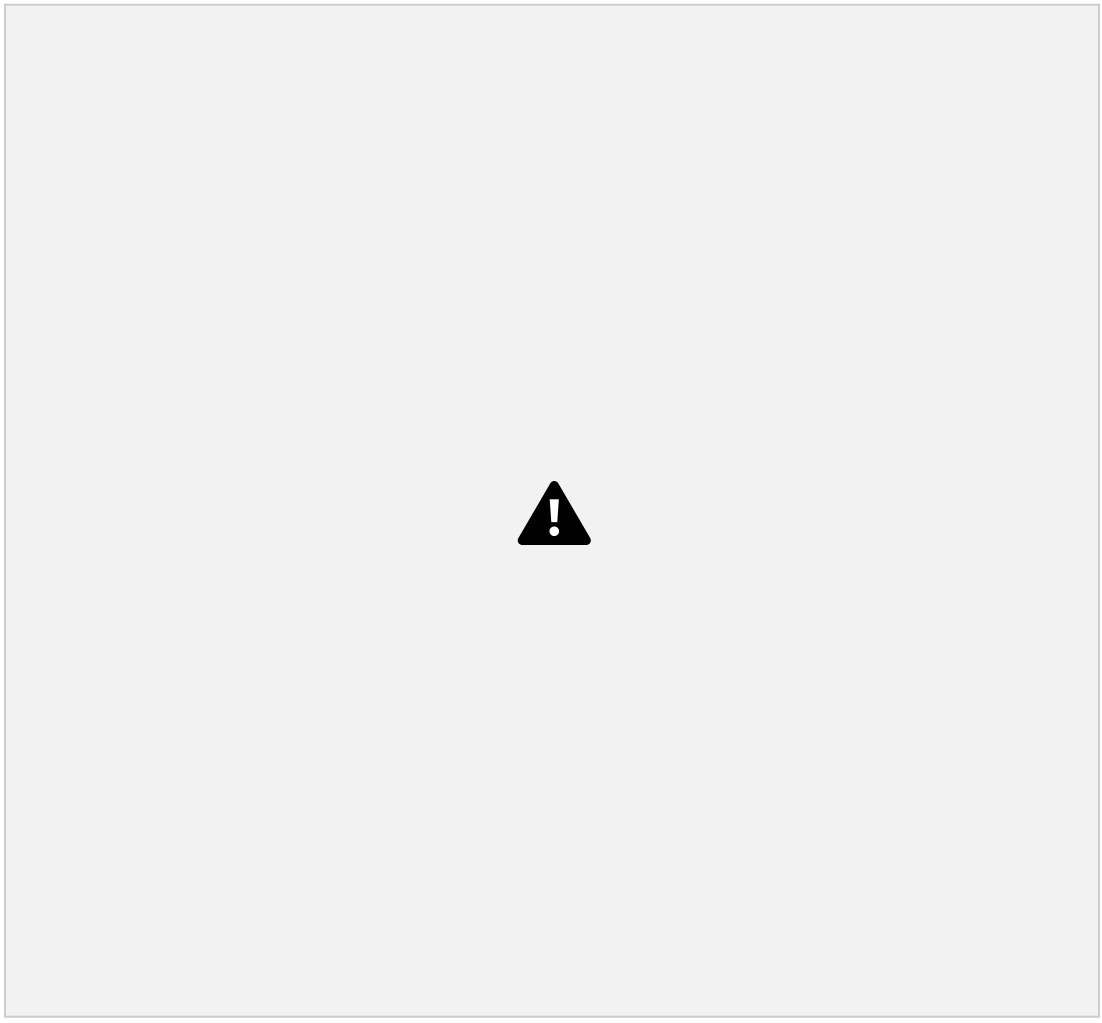
Using the same frequency group are called co-channel cells. The distance between the centers of the nearest co-channel cells, D, relative to the cell radius, R, is given by the co-channel reuse ratio Q:

Co-channel Reuse Ratio Q = D / R = √(3N)

A larger N increases D relative to R (co-channel cells are farther apart), reducing interference but also reducing the number of channels available per cell (since the fixed total spectrum is now divided among more cells in the cluster). This is the fundamental trade-off in cellular frequency planning: capacity vs. interference.

Frequency Reuse Factor:

The fraction of the total available channels used by a single cell is called the frequency reuse factor, equal to 1/N. A smaller cluster size N gives each cell access to a larger share of the spectrum (higher capacity) at the cost of increased co-channel interference, and vice versa.



Algorithm / Procedure

1. Choose a cluster size N and a valid pair of shift parameters (i, j) such that $N = i^2 + ij + j^2$ (e.g., N = 7 with i = 2, j = 1).

2. Generate a hexagonal grid of cells covering a chosen region, using axial coordinates (q, r) to index each cell's position.

3. For every cell (q, r), compute its frequency group using: group = mod(q·j − r·i , N). This assigns each cell a number from 0 to N−1 representing which of the N frequency sets it uses.

4. Convert each cell's axial coordinates to Cartesian (x, y) pixel coordinates for plotting, using the standard hexagon layout formulas.

5. Draw each cell as a hexagon at its computed (x, y) position, filled with a color corresponding to its frequency group, and label it with its group number.

6. Identify and visually verify that cells sharing the same color/group (co-channel cells) are separated by the expected reuse distance D.

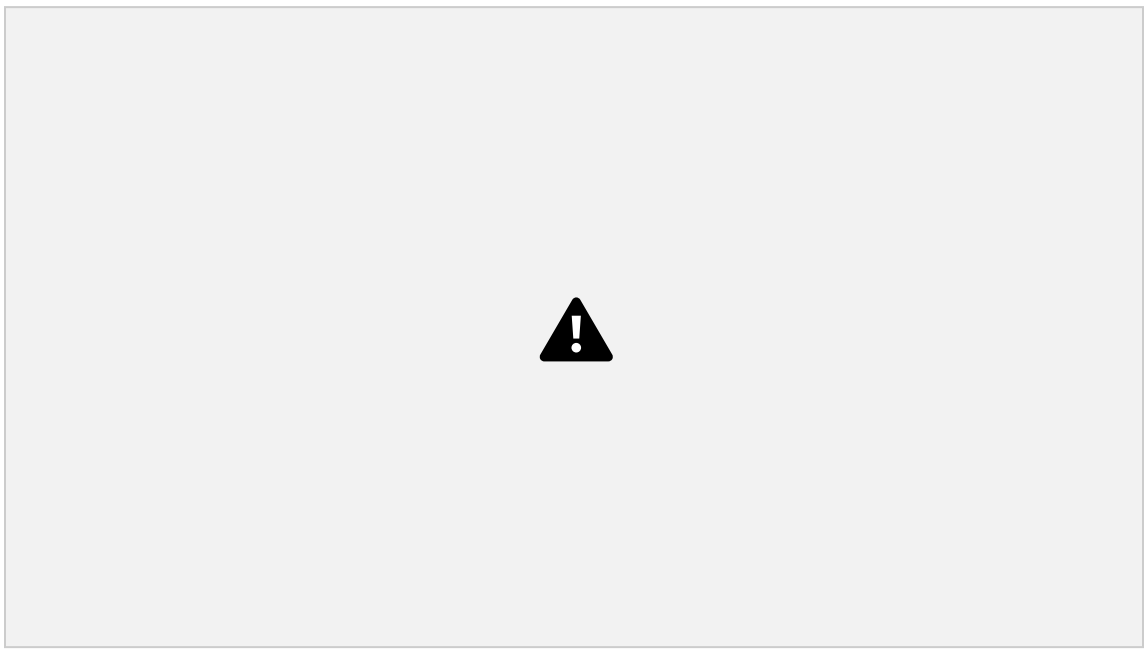
7. Repeat the simulation for a different cluster size (e.g., N = 4 or N = 12) and compare the resulting reuse pattern and spacing between co-channel cells.

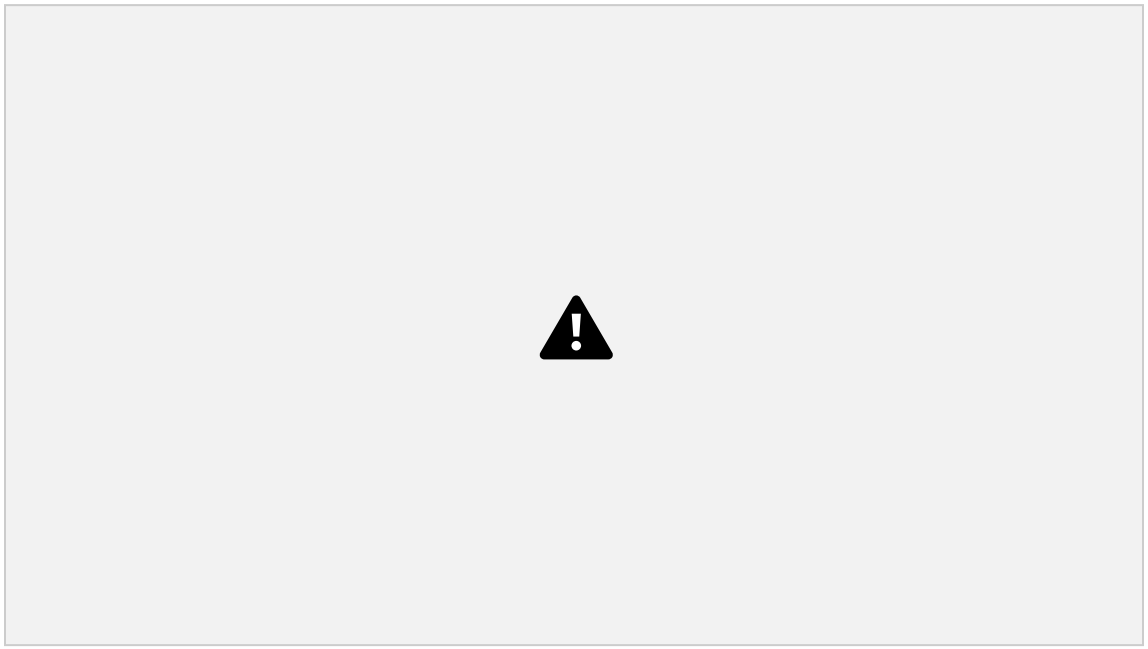
CONCULSION:

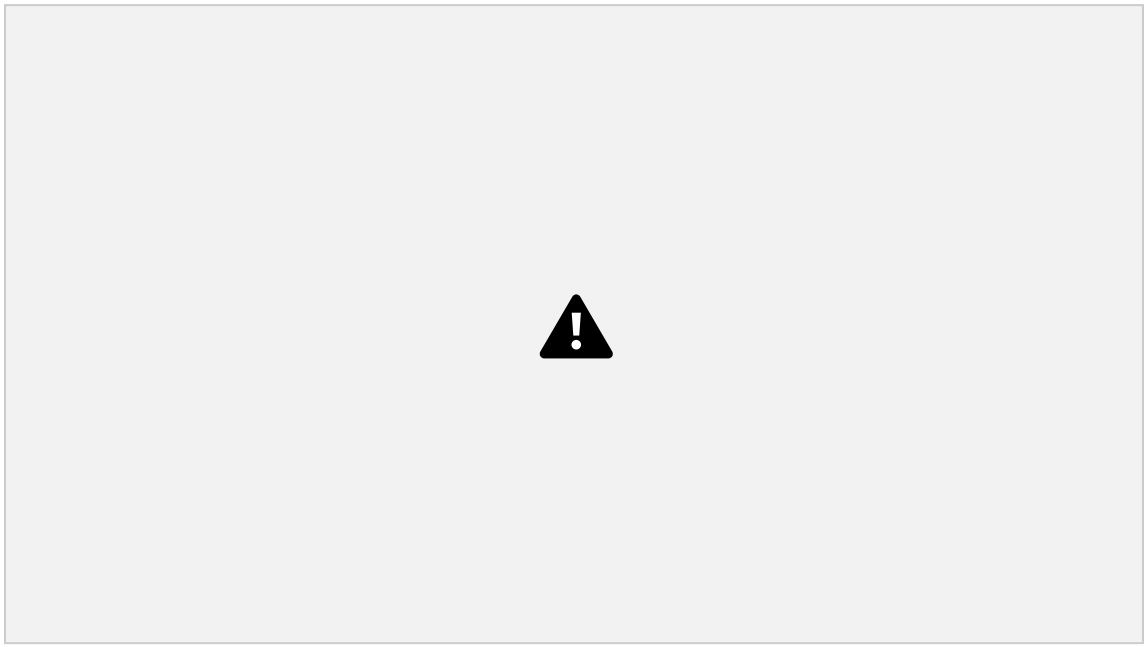
The frequency reuse concept in a cellular communication system was successfully demonstrated using MATLAB. The simulation showed how the same frequency channels can be reused in non-adjacent cells to improve spectrum utilization while minimizing interference. This experiment highlights the importance of frequency reuse in increasing network capacity and ensuring efficient use of the available radio spectrum in modern cellular networks.

OUTPUT:

Code:







Diagram(Output):

